



ONFOODS: A Substitute Recommendation System in Food Recipes

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Abstract. Food waste is a serious problem in modern society. A specific aspect of food waste concerns meat consumption in gastronomy, where typically only prime cuts of meat are used in the kitchen. To facilitate the usage of all parts of animals and thereby reducing food waste, we present ONFOODS, a system that recommends alternative meat cuts in recipes and integrates inventory data to help with the creation of menus. ONFOODS uses an ontology and a knowledge graph to model recipes, meat cuts and the relationships between the two, similarity measures to find candidates for alternative meat cuts, and inventory data to track the availability of different meat cuts. An intuitive user interface allows the user on one hand to update the knowledge graph and inventory data, and on the other hand to navigate through recipes and choose alternative meat cuts.

Keywords: Recommender System · Ingredient Substitution · Ontologies · Knowledge Graphs

1 Introduction

Substituting ingredients in cooking recipes has great potential to improve customer satisfaction and sustainability. For instance, when substituting an ingredient due to dietary restrictions, recipes can be personalized to customer needs. Similarly, when substituting an ingredient in a recipe with an alternative ingredient that may be about to expire or is abundant, food waste can be reduced. Recent studies [1,4,6,17] on ingredient substitution leverage ontologies and knowledge graphs to model and integrate recipe and ingredient data, enabling more accurate substitution recommendations. While these works are motivated by specific goals, such as satisfying user's dietary restrictions, health considerations, or nutritional needs, our motivation is to reduce meat waste. Since recipes require specific meat cuts that may not be at hand, our approach is to combine substitute recommendations with the inventory that indicates which meat cuts are available at which quantities.

Food waste is mentioned in the United Nations' sustainable development goals as a serious problem in modern society. It is estimated that about half of the food grown is wasted before and after it reaches the consumer [14]. A specific instance of this problem is manifested in meat consumption in gastronomy. Instead of using all parts of animals, restaurants often use only the fine cuts of meat, partially because cooks are not familiar with other parts and do not know how to prepare them, and partially because guests consume only fine cuts. One way to help mitigating meat waste in the gastronomy is to provide an information system, which on one hand stores information about the origin of the animal, expiration date of the meat, etc., and on the other hand provides information about how to use alternative meat cuts for the preparation of meals. To develop such an information system, ontologies and knowledge graphs can be used to provide a standardized conceptual terminology [9]. While an ontology represents metadata or a schema, which captures more complex structures with relationships between a set of concepts, the focus of knowledge graphs is data instances. Ontologies and knowledge graphs can be used for various real-world applications such as recommender systems, decision making, and information retrieval.

In this paper, we focus on meat substitution for recipes due to its perishable nature and high cost, making accurate substitution particularly valuable for chefs and other consumers. Currently, most cooks in restaurants and hotels only rely on particular parts of an animal, which are ordered daily from butcher shops. There are several reasons for this: preference for a particular cut, recipes usually only mention one meat cut for a specific recipe, and using an entire animal requires more knowledge about how to use the different parts. To find meat cut alternatives, we use scoring metrics, such as cosine similarity and the Word2Vec model, based on nutritional composition and sensory attributes.

To tackle these problems, we present a recommendation system for alternative meat cuts to be used by chefs and/or decision-makers in restaurants for finding the most sustainable available meat cut for recipes, which can help in considering the use of entire animals. Our system relies on an ontology to organize meat cuts and recipes. The ontology specifies a hierarchy of meat cuts and a hierarchy of recipes together with relationships between the two hierarchies, which allow to find and rank alternative meat cuts for each recipe. Moreover, our system integrates inventory data that helps to manage the inventory and usage of meat.

The technical contributions of this paper can be summarized as follows:

- We design an ontology for meat substitutes in food recipes and generate a knowledge graph based on the ontology.
- We integrate inventory data into the recommendation system, allowing users to create menus based on available meat cuts.
- We investigate how to apply word embeddings and similarity measures based on nutritional composition and sensory attributes to identify suitable alternative meat cuts.

- We provide an intuitive user interface that facilitates easy navigation through the knowledge graph, allowing users to update data and choose alternative meat cuts.

The rest of the paper is structured as follows. In Sect. 2 we discuss and compare related work. Section 3 describes our approach to score alternative meat cuts. In Sect. 4, the architecture of the ONFOODS system is described, including the ontology, knowledge graph, and the inventory database, followed by a discussion of several use cases in Sect. 5.

2 Related Work

Food ontologies specify the shared terminology for types, properties, and relations among food concepts and, hence, can aid in the resolution of data harmonization issues that span food-related domains. Food ontologies are usually used as the schema and play an important role in the development of food knowledge graphs. There exist several food-related ontologies and knowledge graphs: specialized ones in specific domains, e.g., health and nutrition [2, 5], cooking and recipes [7, 19], and food safety [15, 16], as well as more general food ontologies and knowledge graphs [1, 4]. A large-scale and comprehensive knowledge graph is FoodKG [4]. It adopts the food product ontology from FoodOn [1] and covers recipes, ingredients, and nutrition information, which can be used in many applications, such as recipe recommendation and ingredient substitutions.

Several studies have addressed the problem of ingredient substitution in food recipes, employing a variety of techniques, such as heuristics [17], AI techniques [10, 13] semantic models [6], and filtering [12].

The work in [17] presents a heuristic using explicit semantic information and word embeddings of ingredients to rank plausible ingredient substitutions in FoodKG. To rank ingredient substitutions, similarities in the properties of the ingredients and the recipes are used.

The authors of [13] investigated how to use natural language processing techniques such as word embedding to find alternative ingredients or recipes. The proposed system replaces an ingredient with similar ingredients or a recipe with similar recipes using the cosine similarity of the word embeddings of the ingredients. The AdaptaFood system [10] focuses on adapting recipes to dietary needs using AI-driven ingredient extraction and substitution. It employs BERT-based language models and image-captioning techniques to extract and analyze ingredient descriptions, and find semantically potential substitutes. The system identifies substitutes based on how similar two ingredients are in terms of their textual descriptions and contextual usage in recipes.

In [6], the authors introduced an ontology design pattern for food recipe ingredient substitution, aiming to enhance the interoperability and knowledge representation in culinary and nutritional domains. Their approach facilitates the systematic modeling of ingredient alternatives based on various criteria such as nutritional value, flavor compatibility, and dietary restrictions.

The work in [12] proposed an ingredient substitute recommendation system based on a filtering process that takes into consideration the original recipe context, the relationship among sets of ingredients and user preferences. According to a category of user food restrictions, the proposed system replaces forbidden ingredients in recipes with safe ingredients.

While previous approaches focus on technical aspects and offer general ingredient substitution in recipes, in this paper we concentrate on an easy-to-use end-to-end system that suggests alternative meat cuts in recipes with the aim to use entire animals, thereby mitigating food waste. In addition to suggest alternative meat cuts, our system stores inventory data, such as expiry date, the origin of the animal, and the amount of different cuts, which are helpful information to support the optimal use of entire animals.

3 Scoring Meat Cut Alternatives

To support chefs in using all parts of an animal and, hence, reducing food waste, it is important to propose suitable alternative meat cuts for each recipe. Since recipes are typically prepared with specific meat cuts that may not be at hand, our recommendation of alternative meat cuts considers both the quality of the alternatives as well as the inventory that indicates which meat cuts are available at which quantities.

To find and score meat cut alternatives, our system offers two options: First, chefs can add alternative meat cuts for each recipe, based on their experience and preference for flavor and texture. Second, alternative meat cuts are determined based on the similarity of the nutritional compositions (protein, lipid, and water content) and the sensory attributes (tenderness, texture, and flavor) that influence culinary performance. These factors can be helpful in determining the suitability of a substitution since meat cuts with similar nutritional composition may differ significantly in their cooking behavior and sensory attributes. To incorporate these aspects, we use a weighted cosine similarity measure, where different attributes contribute differently to the overall similarity score. The weights w_i are empirically determined based on expert opinions to reflect the relative importance of each factor in substitution decisions. Since some attributes (e.g., protein, lipid, water content) are continuous, while others (e.g., tenderness, texture, flavor) are categorical, the categorical attributes are first mapped into numerical vectors by applying the Word2Vec embedding method [8]. This method captures the semantic similarity between sensory attributes by representing them as dense numerical vectors, which can be directly compared in the similarity computation. Additionally, Min-Max normalization is applied to all attributes, mapping them to a normalized $[0,1]$ range. This ensures a balanced contribution of both nutritional and sensory attributes in the similarity computation. Then, both sensory and nutritional attributes are concatenated into a single feature vector for each meat cut, which is used to compute the cosine similarity. More specifically, a substitution score S between two meat cuts A and B with normalized feature vectors $\langle A_i \rangle$ and $\langle B_i \rangle$ for $i = 1 \dots n$ (representing

the nutritional and sensory attributes) is calculated using the weighted cosine similarity as follows:

$$S(A, B) = \frac{\sum_{i=1}^n w_i A_i B_i}{\sqrt{\sum_{i=1}^n w_i A_i^2} \cdot \sqrt{\sum_{i=1}^n w_i B_i^2}} \quad (1)$$

The weights w_i for each attribute reflect its importance in determining the similarity between two meat cuts. These weights are empirically determined based on expert opinions, ensuring that attributes such as tenderness and flavor are appropriately emphasized alongside nutritional composition.

If alternative meat cuts are used, different quantities might be required for the same recipe, depending mostly on the amount of water in the meat. Thus, to adjust the quantities needed for a recipe, we consider the quantity of dry matter in each meat cut and compute the quantity of meat cut A with respect to the quantity of meat cut B as:

$$Q_A = Q_B \times \frac{100 - \text{Water}_B}{100 - \text{Water}_A} \quad (2)$$

4 System Design and Architecture

The overall system design and architecture of ONFOODS is shown in Fig. 1. It consists of four parts: *substitution module*, *ontology*, *knowledge graph*, and *user interface*. We present in detail the parts of our system in Sects. 4.1 to 4.4. Afterwards, in Sect. 4.5 we describe the implementation of ONFOODS.

4.1 Substitution Process

To find and rank meat cut alternatives, our system offers two options: First, chefs can add alternative meat cuts for each recipe, based on their experience and preference for flavor and texture. Second, alternative meat cuts are determined based on the similarity of the nutritional compositions (protein, lipid, and water content) and the sensory attributes (tenderness, texture, and flavor) that influence culinary performance. As shown in Fig. 1, the substitution process consists of two phases:

1. *Scoring alternatives*: This phase uses scoring metrics, including cosine similarity and the Word2Vec model to automatically suggests alternative meat cuts based on the nutritional compositions and the sensory attributes as discussed in Sect. 3.
2. *Ranking alternatives*: This phase allows the user to refine and customize the suggested alternatives. Users can propose new alternatives beyond those automatically generated by the system, reorder the suggested options, and adjust substitution quantities. This interactive process enhances flexibility by integrating user preferences into the system, improving the adaptability and personalization of substitutions.

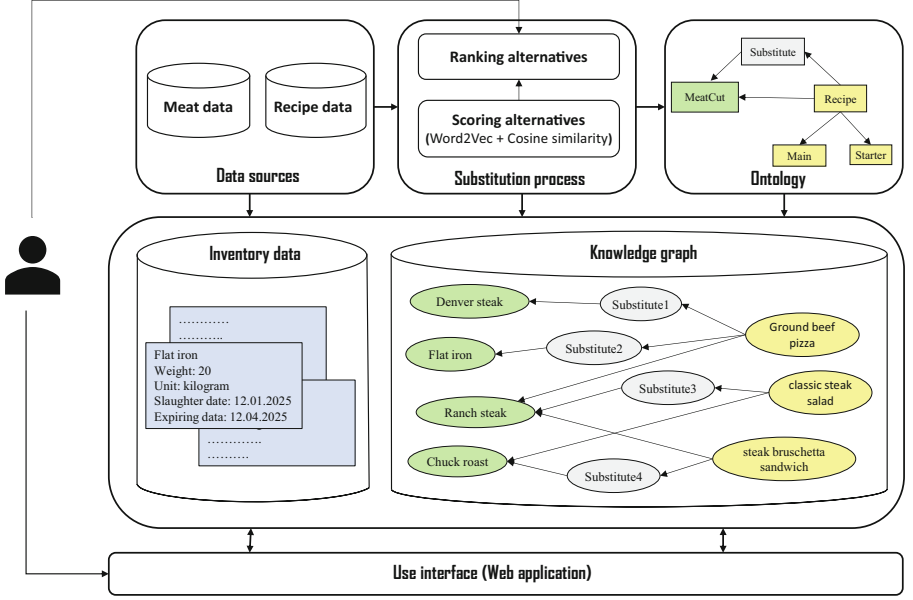


Fig. 1. System architecture.

4.2 Ontology

The ontology specifies the key concepts/classes, relationships and attributes in our specific domain and serves as schema for the knowledge graph. Existing ontologies, such as FoodOn and FoodKG, provide comprehensive models for food items and nutritional information, which only offer generic conceptualizations of substitutions. However, they do not capture the specific substitution relationships required for practical meat cut recommendations, including nutritional and sensory attributes as well as explicit substitution quantities. These features are critical for our use case, where accurate substitutions and the integration with inventory-aware recommendation systems are essential. Thus, the development of a tailored ontology is required.

For the ontology construction, we adopt the method described in [3], which consists of the following five steps: specification, knowledge acquisition, conceptualization, implementation and validation.

- *Specification:* This step entails identifying the purpose of the ontology, defining the scope and objectives of the ontology, and understanding the domain requirements. Our ontology aims at modeling meat cuts and food recipes, and identifying alternative meat cuts to reduce waste.
- *Knowledge acquisition:* This step involves analyzing recipes and meat nutrition information for identifying concepts and relationships, forming an early version of the ontology definition.

- *Conceptualization*: In this step the acquired knowledge is organized into a conceptual model. The identified concepts have been translated into classes, and their attributes have been represented using data properties. The relationships between concepts have been modeled through several object properties.
- *Implementation*: This step uses ontology languages and tools, such as OWL and Protégé [11], to organize classes and properties, define constraints, and implement the ontology.
- *Validation*. The final step ensures correctness and consistency of the ontology using reasoners, such as Pellet [18], an OWL 2 compliant reasoner integrated in Protégé, to evaluate logical integrity and avoid contradictions. Beyond basic consistency checking, we define a set of logical axioms, such as class hierarchies, disjointness, domain and range restrictions, and existential constraints, to formally characterize the classes and relationships in our ontology. This is to ensure not only logical consistency, but also the intended semantic behavior of the ontology for practical applications.

Figure 2 shows the ontology, which models meat data and food recipes with the aim to find alternative meat options for recipes in order to reduce meat waste. The three main classes are *Recipe*, *MeatCut* and *Substitute*. Each recipe is described by several properties, such as quantity, cooking time, and number of servings, and it is recommended for different courses, where we distinguish between starters and main courses. The class *MeatCut* is characterized by nutritional information and sensory attributes. The relationship *hasMeatCut* between the classes *Recipe* and *MeatCut* represents the main cut of meat to be used for a specific recipe. The *Substitute* class represents alternative meat cuts for a recipe, capturing important details, such as the required substitution quantity and a ranking to indicate preferences. The *hasSubstitute* relationship links a recipe to one or more substitute instances, and each substitute instance is further connected to an alternative meat cut through the *isMeatCut* relationship, specifying which meat cut can replace the original. The ontology is publicly available at <https://w3id.org/onfoods/ontology>.

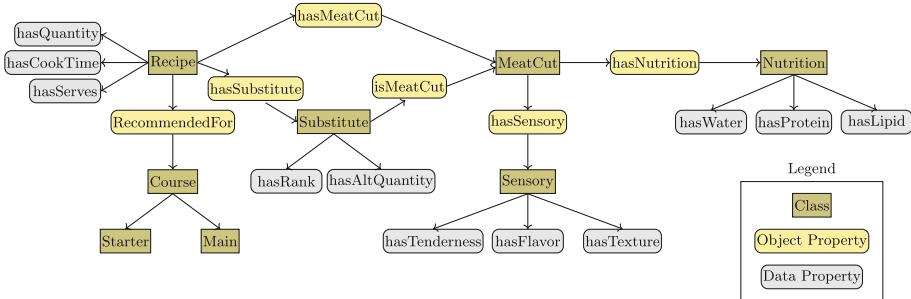


Fig. 2. Ontology

4.3 Knowledge Graph and Inventory Database

The knowledge graph instantiates the schema defined by the ontology with specific instances of the entities and relationships from the domain, i.e., individual recipes, beef cuts, and relationships between these two classes. For the instantiation of the knowledge graph we used two main sources of data:

- The USDA ingredient nutrient database¹, which provides nutritional information for beef cuts, including energy, water content, and macro (carbohydrates, proteins, and lipids) and micro nutrients (vitamins and minerals).
- The second source of data are various online recipe sites.

We chose to collect our recipe data from a variety of sites that provide beef-based recipes, extracting details such as the type of beef cut and its quantity, cooking time, and number of servings. Additionally, users can add individual recipes or adapt existing ones.

As a data storage system for the knowledge graph and inventory data about the availability of different meat cuts, we employ the Open Source database management system PostgreSQL. PostgreSQL does not support graph data structures natively, as Neo4j or other dedicated graph databases do. However, we can achieve this by using PostgreSQL tables to represent nodes and relationships.

4.4 User Interface

To make the system more accessible to a broader audience such as chefs and restaurants managers, we developed a user-friendly web interface for ONFOODS. The main pages are shown in Figs. 3, 4, 5, 6 and 7. The graphical user interface is designed with a focus on usability and accessibility, particularly for non-expert users to interact with the substitution recommendation system without requiring technical knowledge. To ensure ease of use, our design decisions emphasize visual clarity, minimal complexity, and intuitive interactions. Users can input recipes, browse beef cuts, receive substitute recommendations based on our metrics, and rank substitutions based on their experience and preference. The web interface design is based on simplicity with clear input fields, easy-to-navigate options, and results that are easy to interpret.

4.5 Implementation

ONFOODS (<http://onfoods.projects.unibz.it/>) is implemented as a web application, which is built using the ASP.NET MVC framework, leveraging Razor for dynamic web pages and PostgreSQL as the database. The system follows the Model-View-Controller (MVC) architecture, ensuring a clear separation between data management, user interface, and business logic. The backend retrieves available Beef cuts, manages the availability of beef cuts, calculates similarity scores for finding alternative suggestions, and presents them through a responsive

¹ <https://fdc.nal.usda.gov/>.

Razor-based UI. The database schema follows the ontology structure, ensuring semantic organization of the data and making it easier to query and relate different concepts. The frontend of the application is developed using Razor within the ASP.NET MVC framework, ensuring a dynamic and responsive user interface and allowing easy access to functionalities like browsing beef cuts, exploring recipes, and managing inventory.

To address scalability requirements in practical settings, our recommendation system is specifically designed for real-world restaurant environments, where the number of relevant recipes and the granularity of meat cut hierarchies are inherently limited. Specifically, the system supports a three-level hierarchy for meat cuts, reflecting the granularity commonly used in restaurants. During testing we incorporated 200 recipes, which is representative of real-world use cases in restaurants. We observed that the system maintained robust performance and responsiveness as the number of recipes and meat cut variations increased. The system allows for straightforward expansion, making it possible to add new recipes, meat cuts, or substitution options without requiring significant modifications. These features demonstrate that our system is both scalable and adaptable to evolving menu requirements.

5 Use Cases

In our prototype implementation we use beef cuts for a concrete case study. However, the system can easily be applied to other types of meat, such as pork or lamb. In this prototype, due to the availability of data, we primarily rely on nutritional attributes with equal weighting to determine similarity, as sensory attributes for beef cuts were not readily available. In the following we showcase various scenarios that illustrate how users can interact with the ONFOODS system (cf. Figs. 3, 4, 5, 6 and 7).

Scenario 1 – Browsing Beef Cuts. Figure 3 shows how users can explore a variety of beef cuts through the hierarchical structure of beef parts, check their availability in the inventory, and browse through various recipes that use this particular beef cut. This functionality is particularly useful when creating menus because the availability is considered to reduce food waste.

Scenario 2 – Alternative Beef Cuts. This scenario highlights the application’s ability to suggest alternative beef cuts based on similar characteristics and the chef’s preferences. In Fig. 4, users see the main beef cut for a selected recipe, i.e., a ranked list of recommended alternatives together with cooking time, course, and serves. Each beef cut has assigned two values, the required quantity for the recipe and the selected servings (adjusted based on Sect. 3) and its availability in the inventory. Users can select a beef cut based on the availability. If a beef cut is selected, the inventory is updated prioritizing cuts with earlier expiration date. Consider Fig. 4, which shows two alternative beef cuts for 1.30 kg of “Back ribs”: “Large end” is the most similar beef cut, and 1.50 kg are needed. “Short

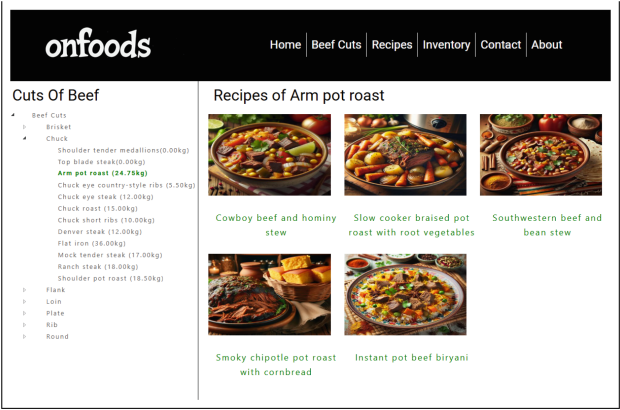


Fig. 3. Hierarchy of beef cuts

ribs” is the second most similar beef cut, for which 1kg are needed for the same recipe.

Scenario 3 – Managing Recipes. The recipes page in Fig. 5a allows users to edit and manage recipes by browsing the list, viewing recipe details, deleting existing recipes, or adding new ones. Figure 5b shows how to add a new recipe. After adding a recipe, the user is redirected to a page that suggests alternative beef cuts based on the similarity score discussed in Sect. 3 (Fig. 6). Additionally, the system allows users to manually add alternative beef cuts and specify their quantities based on their experience, as well as preferences for flavor and texture. Users can also set a ranked order for alternative options and adjust the quantities suggested by the system. Consider Fig. 6, which presents three

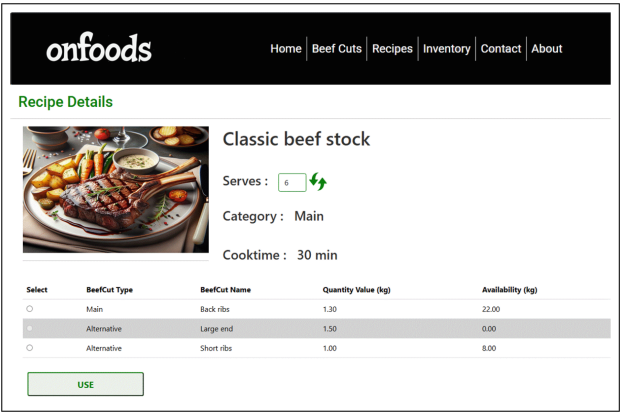
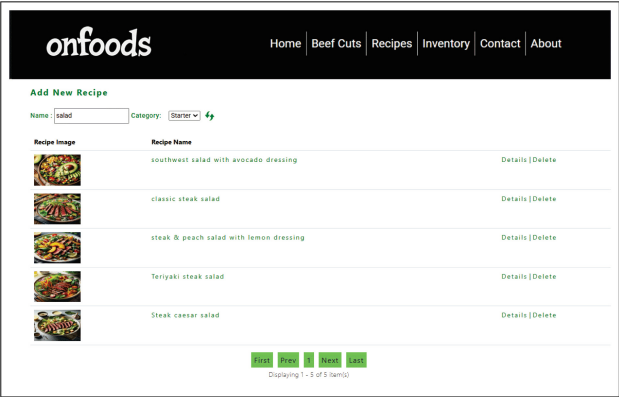
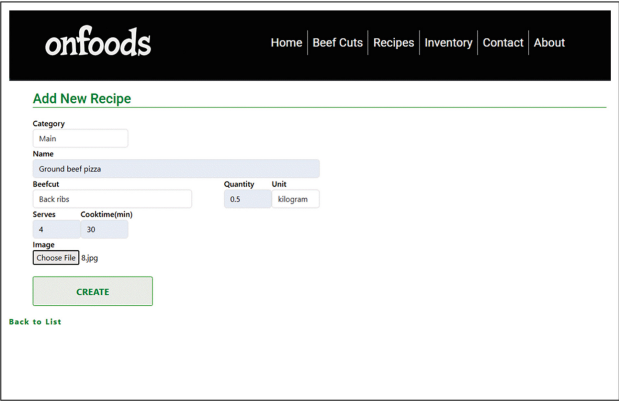


Fig. 4. Details of recipe

alternative beef cuts for the recipe “Ground beef pizza”. In addition to the system-suggested alternatives “Large end” and “Short ribs”, the user proposes an additional option “Flat iron”. The user can also set a ranked order for the alternatives and adjusts their quantities.



(a) Recipes



(b) Adding new recipes

Fig. 5. Managing recipes

Scenario 4 – Managing Inventory. The Inventory page in Fig. 7 allows users to manage the availability of beef cuts. This scenario demonstrates how chefs or decision-makers can monitor the inventory and add newly purchased items. By integrating inventory management with the recommendation system, the system ensures that menu planning aligns with sustainability goals while prioritizing the use of soon-to-expire beef cuts, thereby minimizing food waste and ensuring efficient stock usage.

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Create New Alternative BeefCut

Recipe

Ground beef pizza

Select new Alternative BeefCut

Shoulder tender medallions

Quantity (kg)

0.87

Add

Main BeefCut	Quantity (kg)	Water (g)	Nutrient Data (Portion: 100 gr)	
			Protein (g)	Lipid(g)
Back ribs	0.50	71.50	21.20	6.59

Rank	Alternative Beefcuts	Quantity (kg)	Water (g)	Nutrient Data (Portion: 100 gr)	
				Protein (g)	Lipid(g)
3	Large end	0.30	53.00	16.00	29.30
2	Short ribs	0.28	48.30	14.40	36.20
1	Flat iron	0.70	70.00	18.80	11.30

Save

Fig. 6. Adding alternatives

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Add New Item into Inventory

Beef cut:

Beef cut	Slaughter date	Expiry date	Weight	Unit
Arm pot roast	1/7/2025 12:00:00 AM	8/15/2025 12:00:00 AM	20.00	kilogram
Flat iron	11/7/2024 12:00:00 AM	7/30/2025 12:00:00 AM	8.00	kilogram
Flank steak	1/6/2025 12:00:00 AM	7/20/2025 12:00:00 AM	10.00	kilogram
Denver steak	11/7/2024 12:00:00 AM	8/16/2025 12:00:00 AM	12.00	kilogram
Ribeye cap steak	1/7/2025 12:00:00 AM	8/15/2025 12:00:00 AM	8.00	kilogram
Back ribs	1/7/2025 12:00:00 AM	7/31/2025 12:00:00 AM	10.00	kilogram
Ranch steak	1/28/2025 12:00:00 AM	8/30/2025 12:00:00 AM	6.00	kilogram
Ranch steak	1/31/2025 12:00:00 AM	8/1/2025 12:00:00 AM	12.00	kilogram
Chuck roast	11/28/2024 12:00:00 AM	7/28/2025 12:00:00 AM	15.00	kilogram
Brisket flat	10/25/2024 12:00:00 AM	7/25/2025 12:00:00 AM	9.00	kilogram

First

Prev

1

2

3

Next

Last

Displaying 1 - 10 of 25 items

Fig. 7. Inventory data

6 Conclusion and Future Work

Our paper presents ONFOODS, a recommendation system for alternative meat cuts that assists chefs and/or decision-makers in restaurants to find the most sustainable alternative meat cuts for recipes and manage the inventory and usage of meat. ONFOODS relies on an ontology to model recipes, meat cuts and the relationships between the two. To provide an efficient way for users to interact with the system, we developed a user-friendly web application for ONFOODS with a focus on visual clarity, minimal complexity, and intuitive interactions.

In the future, we plan to enhance the system by integrating machine learning algorithms to improve the accuracy of meat cut recommendations based on historical usage patterns and chef preferences. Additionally, collaborations with local chefs are planned, which will help us to suggest region-specific alternative meat cuts that align with local preferences and sustainability goals.

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